

- 7 Fig. 7.1 shows a circuit containing a capacitor of capacitance C and a resistor of resistance R .

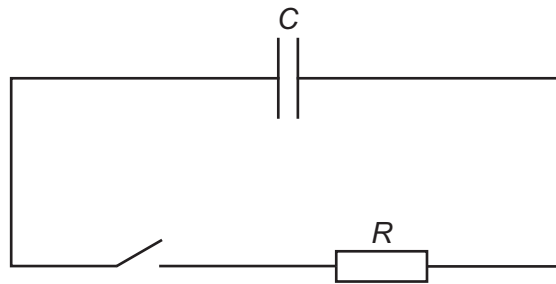


Fig. 7.1

Initially, the switch is open and the potential difference (p.d.) across the capacitor is 12 V.

The switch is closed at time $t = 0$ and the capacitor discharges through the resistor.

Fig. 7.2 shows the variation of the charge Q on the capacitor with the p.d. V_C across the capacitor as the capacitor discharges. Fig. 7.3 shows the variation of the current I in the resistor with the p.d. V_R across the resistor as the capacitor discharges.

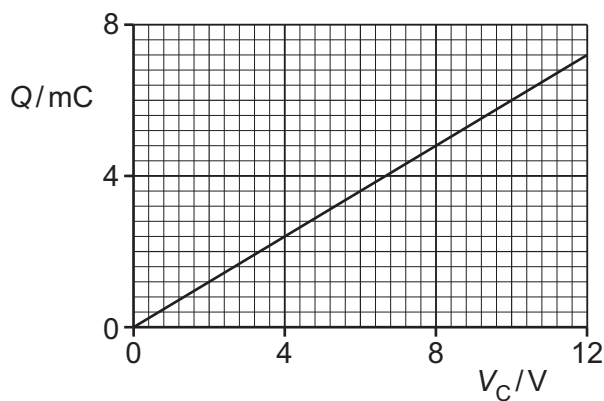


Fig. 7.2

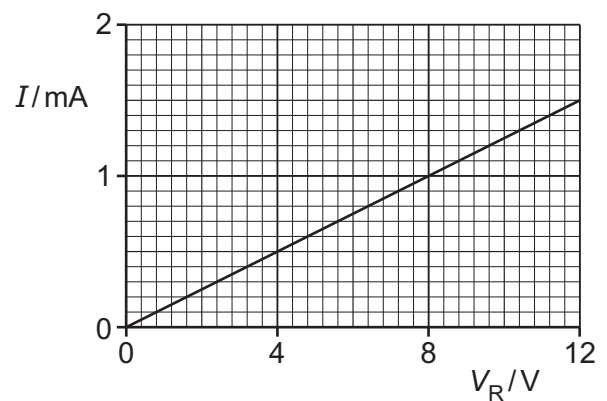


Fig. 7.3

- (a) State the relationship between V_C and V_R .

..... [1]

- (b) Determine:

- (i) the capacitance C , in μF

$C = \dots\dots\dots \mu\text{F}$ [2]

